

# Green Chemistry and Polymers

# Definition of green chemistry

Green chemistry is the design of chemical products and processes that reduce or eliminate the use or generation of hazardous substances. Green chemistry applies across the life cycle of a chemical product, including its design, manufacture, use, and ultimate disposal.

Green chemistry:

- Prevents pollution at the molecular level
- Is a philosophy that applies to all areas of chemistry, not a single discipline of chemistry
- Applies innovative scientific solutions to real-world environmental problems
- Results in [source reduction](#) because it prevents the generation of pollution
- Reduces the negative impacts of chemical products and processes on human health and the environment
- Lessens and sometimes eliminates hazard from existing products and processes
- Designs chemical products and processes to reduce their intrinsic hazards

# Green chemistry's 12 principles

Developed by **Paul Anastas and John Warner in 1998**, the following list outlines a framework for making a greener chemical, process, or product.

These principles demonstrate the breadth of the concept of green chemistry:

- 1. Prevent waste:** Design chemical syntheses to prevent waste. Leave no waste to treat or clean up.
- 2. Maximize atom economy:** Design syntheses so that the final product contains the maximum proportion of the starting materials. Waste few or no atoms.
- 3. Design less hazardous chemical syntheses:** Design syntheses to use and generate substances with little or no toxicity to either humans or the environment.
- 4. Design safer chemicals and products:** Design chemical products that are fully effective yet have little or no toxicity.
- 5. Use safer solvents and reaction conditions:** Avoid using solvents, separation agents, or other auxiliary chemicals. If you must use these chemicals, use safer ones.
- 6. Increase energy efficiency:** Run chemical reactions at room temperature and pressure whenever possible.

# Green chemistry's 12 principles

**7. Use renewable feedstocks:** Use starting materials (also known as feedstocks) that are renewable rather than depletable. The source of renewable feedstocks is often agricultural products or the wastes of other processes; the source of depletable feedstocks is often fossil fuels (petroleum, natural gas, or coal) or mining operations.

**8. Avoid chemical derivatives:** Avoid using blocking or protecting groups or any temporary modifications if possible. Derivatives use additional reagents and generate waste.

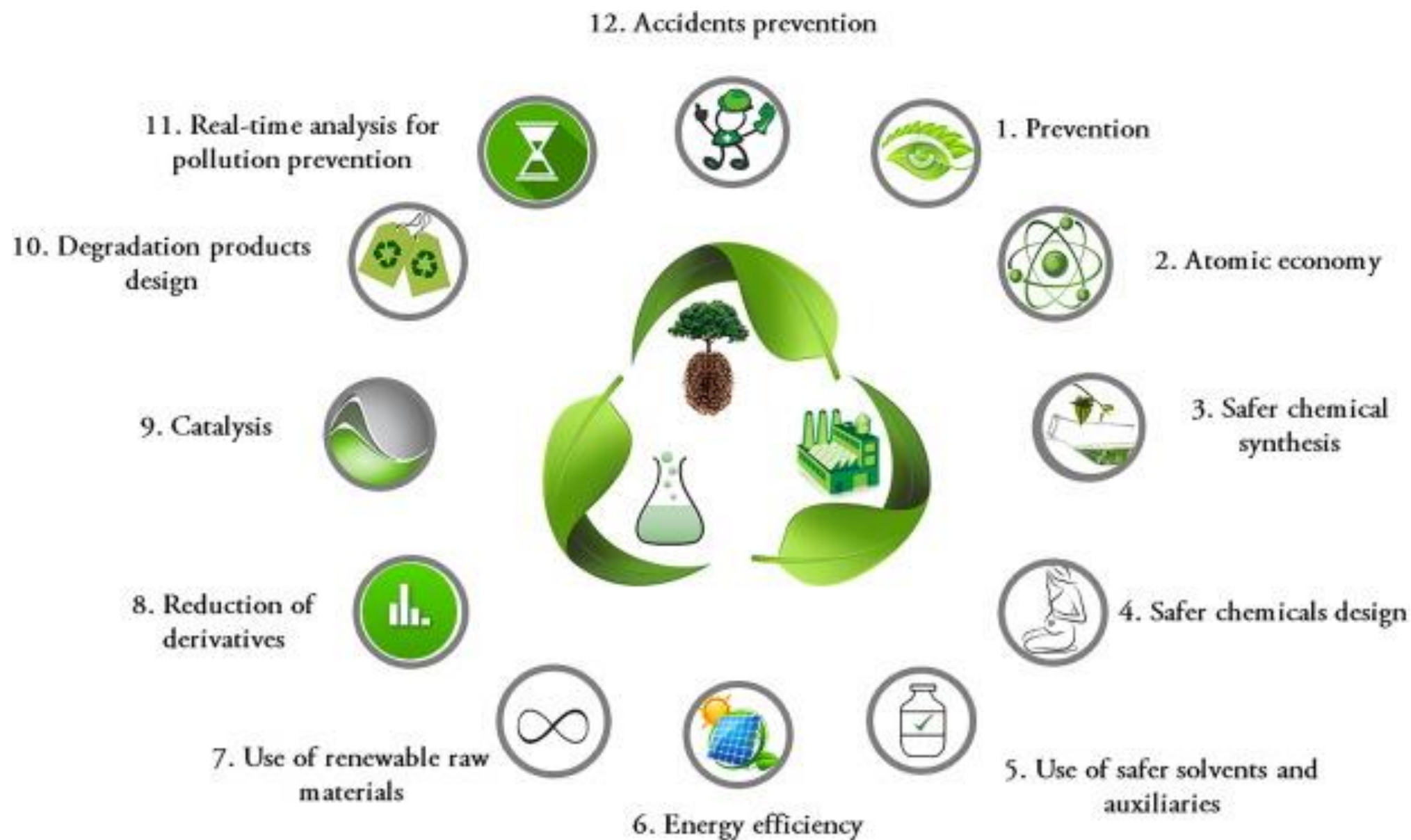
**9. Use catalysts, not stoichiometric reagents:** Minimize waste by using catalytic reactions. Catalysts are effective in small amounts and can carry out a single reaction many times. They are preferable to stoichiometric reagents, which are used in excess and carry out a reaction only once.

**10. Design chemicals and products to degrade after use:** Design chemical products to break down to innocuous substances after use so that they do not accumulate in the environment.

**11. Analyze in real time to prevent pollution:** Include in-process, real-time monitoring and control during syntheses to minimize or eliminate the formation of byproducts.

**12. Minimize the potential for accidents:** Design chemicals and their physical forms (solid, liquid, or gas) to minimize the potential for chemical accidents including explosions, fires, and releases to the environment.

# GREEN CHEMISTRY PRINCIPLES



# Prevention of Waste/By-Products

It is better to **prevent waste than to treat or clean up** waste after it has been created.

- carry out a synthesis in such a way so that formation of **waste (by products) is minimum or absent**.
- cost involved in **the treatment and disposal of waste** adds to the overall production cost.
- the **unreacted starting materials** also form part of the waste. if discharged causes pollution and requires expenditure for cleaning-up.

**Maximum Incorporation** of the Reactants into the Final Product/ **Atom economy**

Synthetic methods should be designed to **maximize the incorporation of all materials** (starting materials and reagents) used in the process into the final product.

**Atom economy** is one of the fundamental principles of green chemistry. Atom economy looks **at the number of atoms in the reactants** that end up in the final product.

$$\% \text{ atom economy} = \frac{\text{Mass of desired product(s)}}{\text{Total mass of reactants}} \times 100$$

# Atom economy

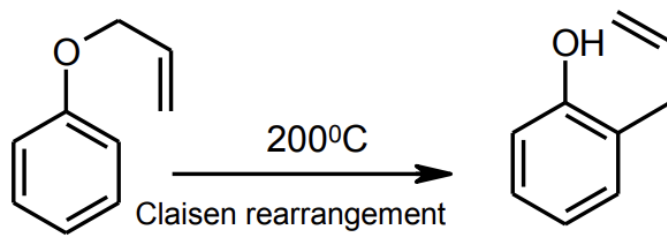
The **atom economy** of a reaction depends on the reagents used in the particular reaction and the type of chemical reaction involved.

- Most of common chemical reactions can be classified as **rearrangement** (e.g migration of an alkyl group), **addition**, **substitution** (e.g chlorination of methane) or **elimination** (e.g dehydration).
- **Rearrangement and addition** reactions are **atom economical** by their very nature, since they simply involve repositioned of reactant atoms within the same molecule or incorporated within a second molecule.
- **Substitution reactions**, however, involve replacement of one group with another and therefore have intrinsically **poor atom economy**.
- **Elimination reactions** is also inherently atom **uneconomical** because of the fact that eliminated atoms are always **lost as waste**.

In developing an atom economical reaction pathway, therefore, the **industrial chemist** may well prefer **rearrangement and addition reactions** over less environmental friendly substitution and elimination reactions.

## 1. Rearrangement Reactions:

These reactions involves rearrangement of atoms that forms molecule. Hence, the atom economy of these reactions are 100%.

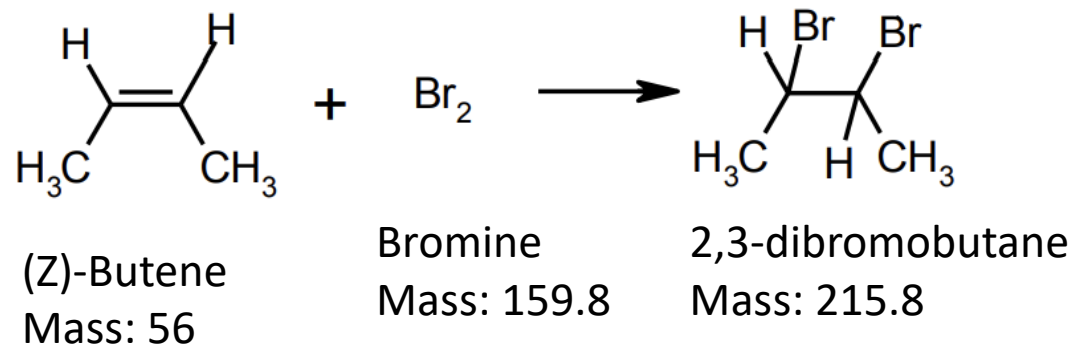


In this rearrangement reaction, % Atom economy =  $(134.175/134.175) \times 100 = 100\%$

# Addition Reaction

## 2. Addition reaction: – halogenation of an alkene

a)



Total Mass of reactants = 56 + 159.8 = 215.8

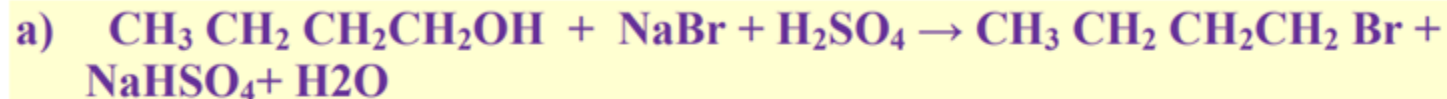
Total Mass of Product = 215.8

% atom economy =  $215.8/215.8 \times 100 = 100\%$

This process is 100% atom efficient,  
with all the reactant atoms included  
within the desired product.

# Substitution Reaction

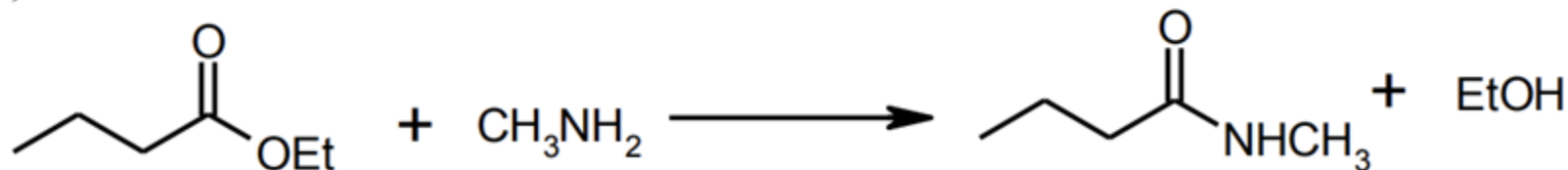
3. **Substitution reaction:** One or more atoms substituted by another atom or group of atoms.



Here, -OH group substituted by a - Br group

$$\begin{aligned}\% \text{ Atom Economy} &= \frac{\text{Mass Of (4C + 9H + 1Br)atoms}}{\text{Mass of (4C + 12H + 5O + 1Br + 1Na + 1S)atoms}} \times 100 \\ &= \frac{137\text{amu}}{275\text{amu}} \times 100 = 50\%\end{aligned}$$

b)

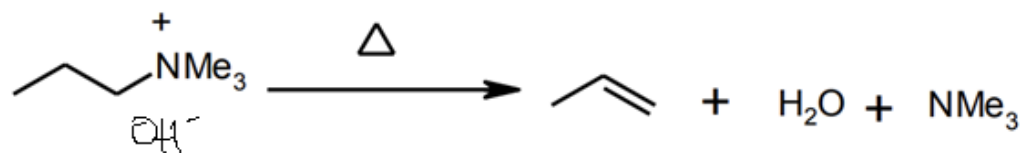


$$\% \text{ Atom Economy} = (87.120 / 133.189) \times 100 = 65.41 \%$$

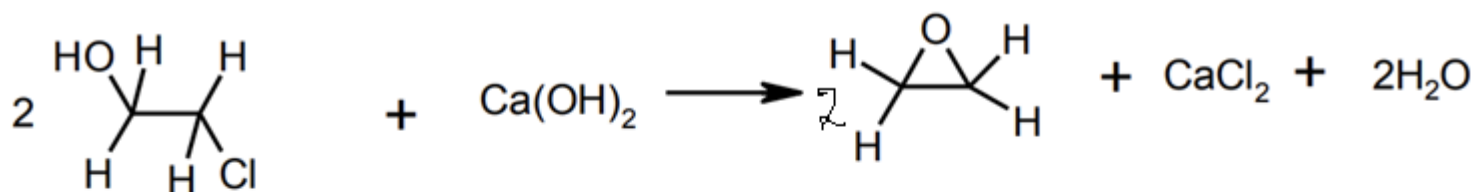
# Elimination Reaction

## 4. Elimination reaction:

A)



$$\% \text{ atom economy} = (42.080 / 119.205) \times 100 = 35.30 \%$$



$$2[(12 \times 2) + (5 \times 1) + 16 + 35.5] \quad 40 + 2(16 + 1) \quad 2[(12 \times 2) + (4 \times 1) + 16] \quad 40 + (2 \times 35.5) \quad 2 \times 18$$

$$= 161 \text{ g}$$

$$= 74 \text{ g}$$

$$= 88 \text{ g}$$

$$= 111 \text{ g}$$

$$= 36 \text{ g}$$

Total mass of reactants = 161 g + 74 g = 235 g

(Note: Total product mass = 235 g)

Mass of desired product ethylene oxide = 88 g

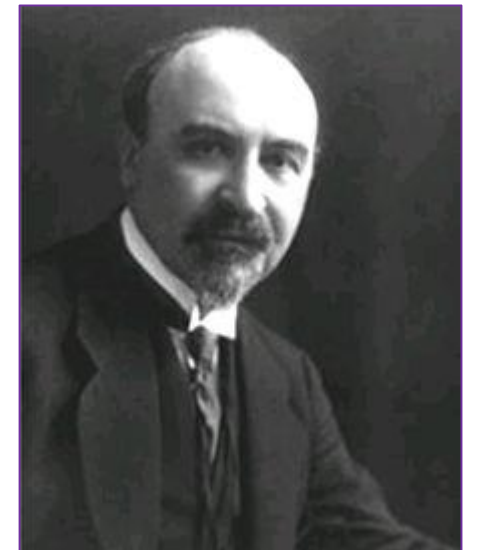
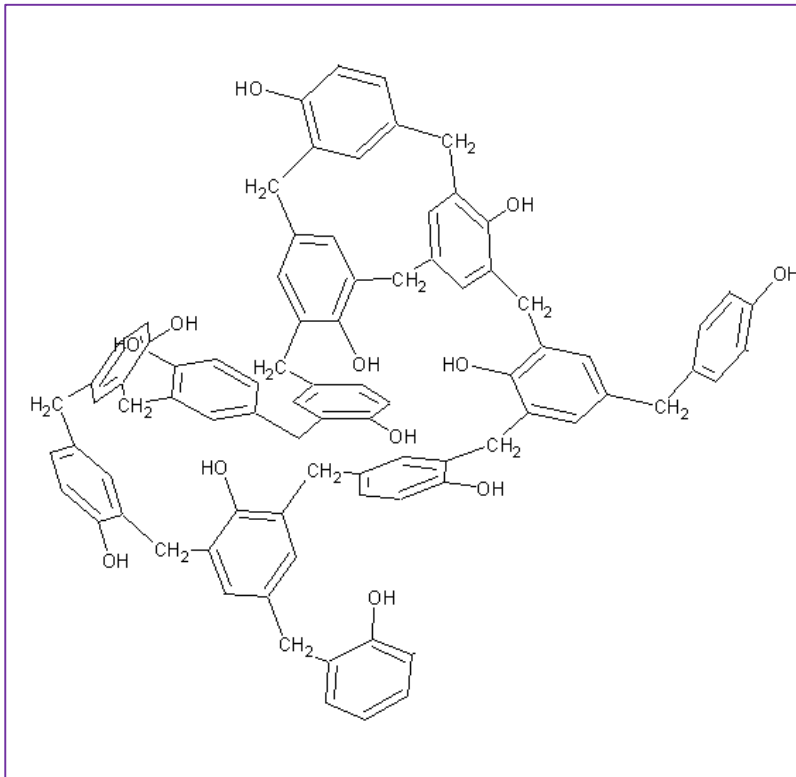
$$\% \text{ Atom economy} = 88/235 \times 100 = 37.4\%$$

# Macromolecules (Polymers)

# Polymers

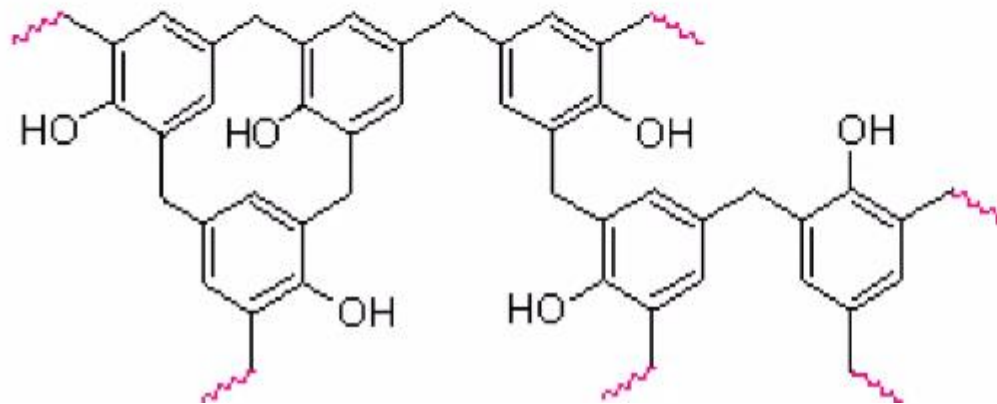
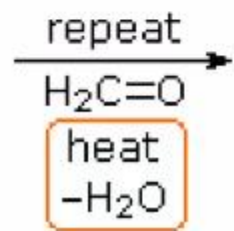
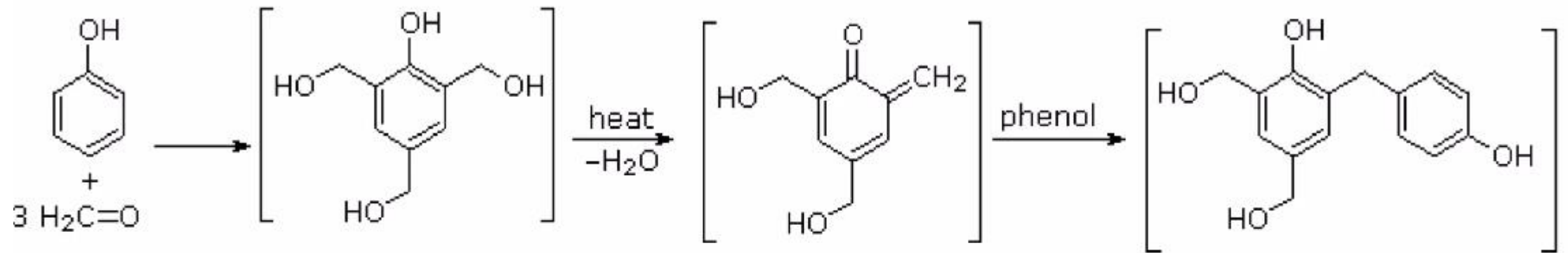
**Bakelite**, formally **Polyoxybenzylmethyleneglycolanhydride**, is a [thermosetting phenol formaldehyde resin](#), formed from a [condensation reaction](#) of [phenol](#) with [formaldehyde](#). The first [plastic](#) made from synthetic components, it was developed by [Leo Baekeland](#) in [Yonkers, New York](#) in 1907, and patented on December 7, 1909.

Because of its electrical [nonconductivity](#) and heat-resistant properties, it became a great commercial success. It was used in electrical insulators, radio and telephone casings, and such diverse products as kitchenware, jewelry, pipe stems, children's toys, and firearms.



## Bakelite

### Phenol-Formaldehyde Resin



**Bakelite** (a cross-linked solid)

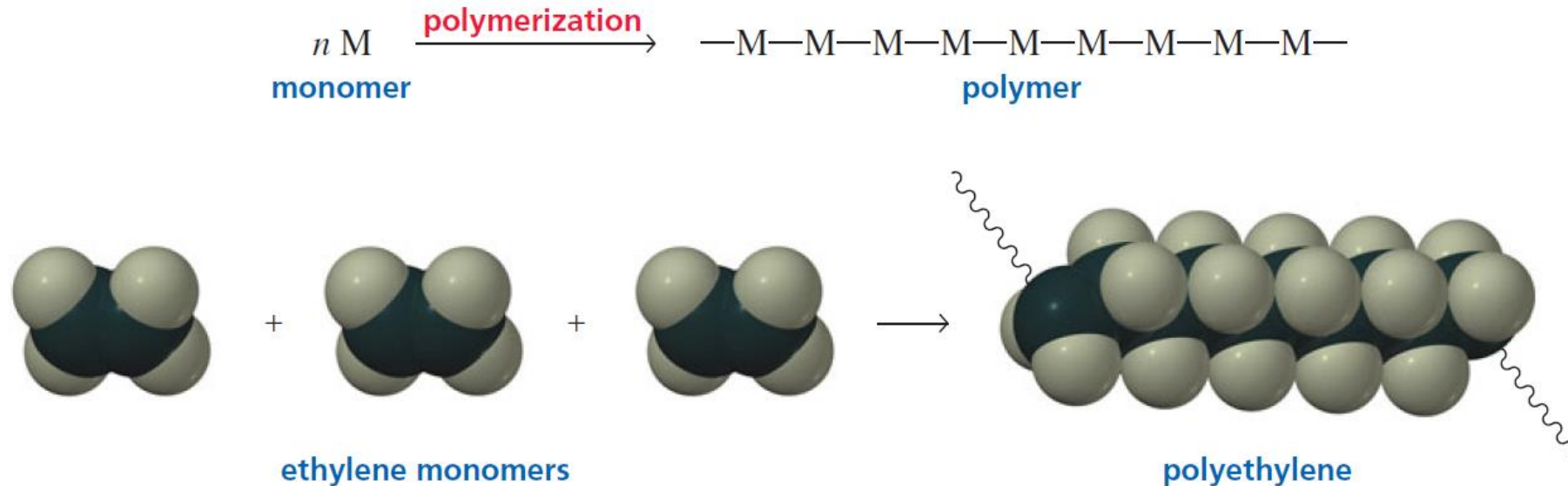
# Polymers

A WORD POLYMER IS A COMBINATION OF TWO GREEK WORDS, "POLY" MEANS "MANY" AND "MEROS" MEANING "PARTS OR UNITS".

- Plastics have replaced an increasing number of natural substances
- Synthetic textile fibers have revolutionized the textile industry
- the low cost, light weight, strength and design adaptability of plastics meet a variety of applications
- Nylon replaced silk in military applications such as parachute and flak vests, and was used in many types of vehicle tires
- A synthetic fiber manufacturing facility occupies a much smaller area of ground than would be needed to produce an equal quantity of natural fibers, such as cotton, wool or silk.

# Polymers

A **polymer** is a large molecule made by linking together repeating units of small molecules called **monomers**. The process of linking them together is called **polymerization**.

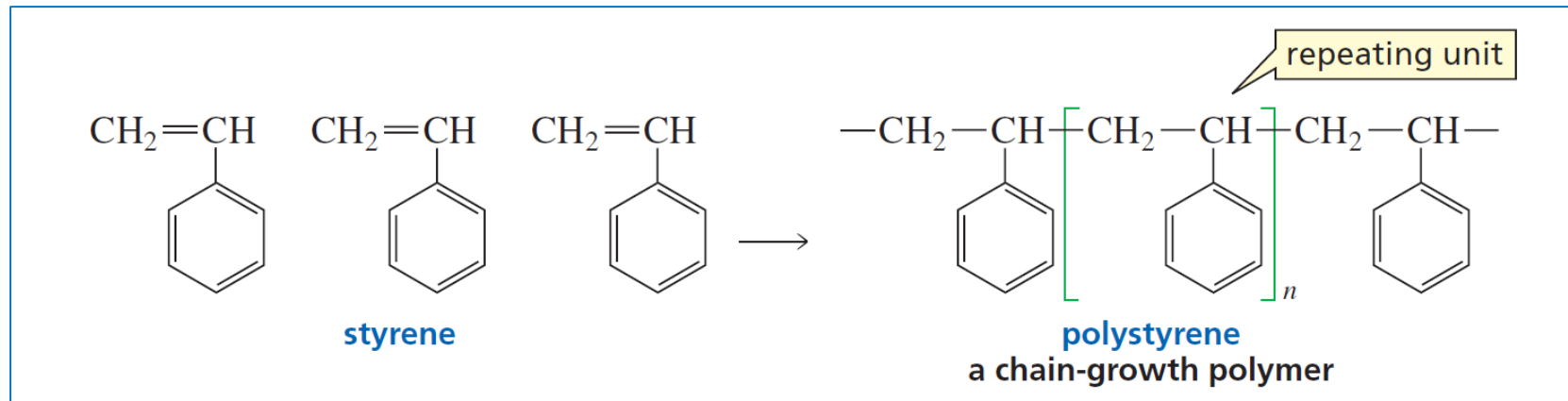


Polymers can be divided into two broad groups: **synthetic polymers** and **biopolymers**. Synthetic polymers are synthesized by scientists, whereas biopolymers are synthesized by cells. Examples of biopolymers are DNA—the storage molecule for genetic information; RNA and proteins—the molecules that facilitate biochemical transformations; and polysaccharides—compounds that store energy and also function as structural materials.

# Polymers

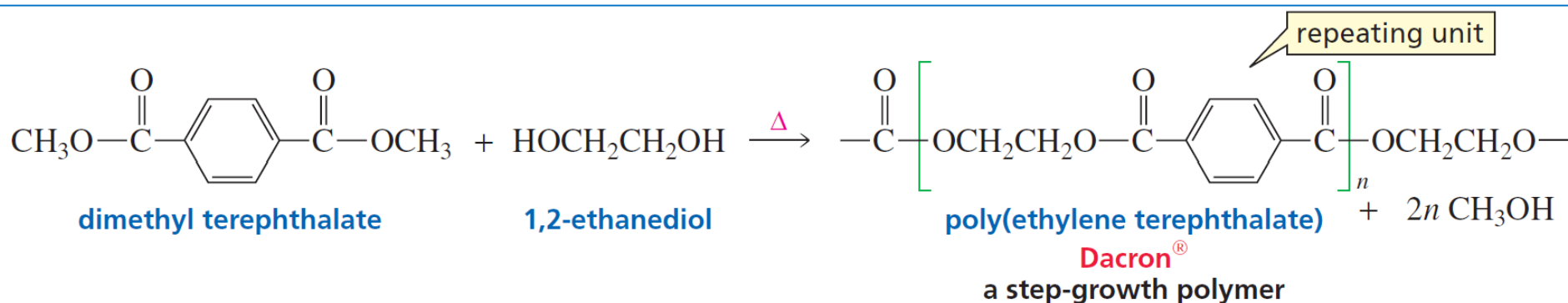
## THERE ARE TWO MAJOR CLASSES OF SYNTHETIC POLYMERS

Synthetic polymers can be divided into two major classes. **Chain-growth polymers**, also known as **addition polymers**, are made by **chain reactions**—the addition of monomers to the end of a growing chain. The end of a growing chain is reactive because it is a radical, a cation, or an anion. Polystyrene—used for hot drink cups, egg cartons, and insulation, among other things—is an example of a chain-growth polymer. Polystyrene is pumped full of air to produce the material used as insulation in house construction.



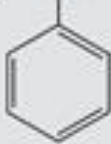
# Polymers

**Step-growth polymers**, also called **condensation polymers**, are made by linking monomers as a result of removing (in most cases) a small molecule, generally water or an alcohol. The monomers have reactive functional groups at each end. Unlike chain-growth polymerization, which requires the individual monomers to add to the end of a growing chain, step-growth polymerization allows any two reactive monomers or oligomers to be linked. Dacron is an example of a step-growth polymer. It was one of the first synthetic polymers to have a medical application—namely, for arterial grafts.



## CHAIN-GROWTH POLYMERS

The monomers used most commonly in chain-growth polymerization are ethylene (ethene) and substituted ethylenes ( $\text{CH}_2=\text{CHR}$ ). Polymers formed from ethylene or substituted ethylenes are called **vinyl polymers**. Some of the many vinyl polymers synthesized by chain-growth polymerization are listed in Table

| Monomer                                                                                                                                         | Repeating unit                                                                                                                                    | Polymer name                        | Uses                                                              |
|-------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|-------------------------------------------------------------------|
| $\text{CH}_2=\text{CH}_2$                                                                                                                       | $-\text{CH}_2-\text{CH}_2-$                                                                                                                       | polyethylene                        | toys, water bottles, grocery bags                                 |
| $\text{CH}_2=\underset{\text{Cl}}{\text{CH}}$                                                                                                   | $-\text{CH}_2-\underset{\text{Cl}}{\text{CH}}-$                                                                                                   | poly(vinyl chloride)                | shampoo bottles, pipe, siding, flooring, clear food packaging     |
| $\text{CH}_2=\text{CH}-\text{CH}_3$                                                                                                             | $-\text{CH}_2-\underset{\text{CH}_3}{\text{CH}}-$                                                                                                 | polypropylene                       | molded caps, margarine tubs, indoor/outdoor carpeting, upholstery |
| $\text{CH}_2=\underset{\text{C}_6\text{H}_5}{\text{CH}}$<br> | $-\text{CH}_2-\underset{\text{C}_6\text{H}_5}{\text{CH}}-$<br> | polystyrene                         | compact disc jackets, egg cartons, hot drink cups, insulation     |
| $\text{CF}_2=\text{CF}_2$                                                                                                                       | $-\text{CF}_2-\text{CF}_2-$                                                                                                                       | poly(tetrafluoroethylene)<br>Teflon | nonstick surfaces, liners, cable insulation                       |

## CHAIN-GROWTH POLYMERS

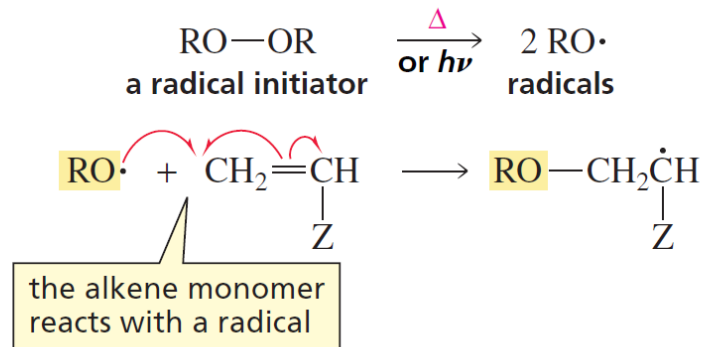
Chain-growth polymerization proceeds by one of three mechanisms: **radical polymerization**, **cationic polymerization**, or **anionic polymerization**. Each mechanism has **three distinct** phases: an ***initiation step*** that starts the polymerization, ***propagation steps*** that allow the chain to grow, and ***termination steps*** that stop the growth of the chain.

### Radical Polymerization

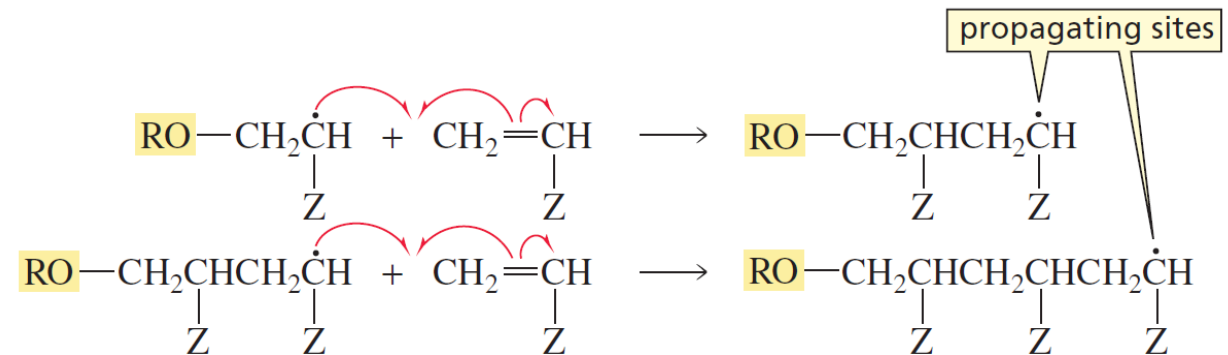
*Radical polymerization* consists of **chain-initiating**, **chain-propagating**, and **chain terminating steps**

#### MECHANISM FOR RADICAL POLYMERIZATION

initiation steps



propagation steps



## Radical Polymerization

- The initiator breaks homolytically into radicals, and each of those radicals can react with a monomer.
- In the first propagation step, the monomer radical reacts with another monomer, converting it into a radical.
- This radical reacts with another monomer, adding a new subunit to the chain. The unpaired electron is now at the end of the unit most recently added to the chain. This is called the **propagating site**.

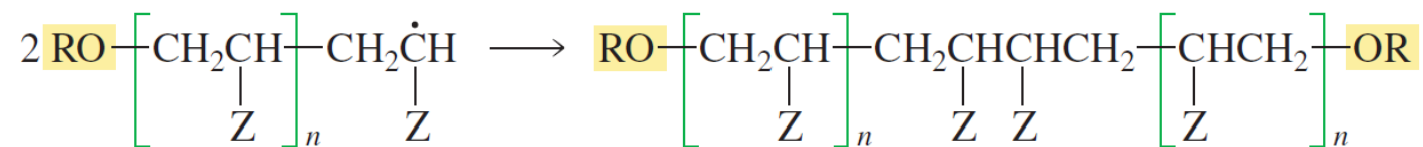
The propagation steps are repeated over and over. Hundreds or even thousands of alkene monomers can add one at a time to the growing chain. Eventually, the chain reaction stops because the propagating sites are destroyed in a termination step. Propagating sites are destroyed when

- two chains combine at their propagating sites.
- two chains undergo *disproportionation*, with one chain being oxidized to an alkene and the other being reduced to an alkane as a result of hydrogen atom being transferred from one chain to another.

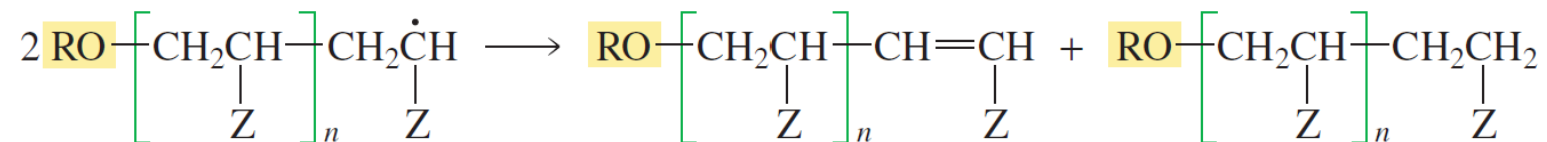
## Radical Polymerization

### termination steps

#### chain combination



#### disproportionation

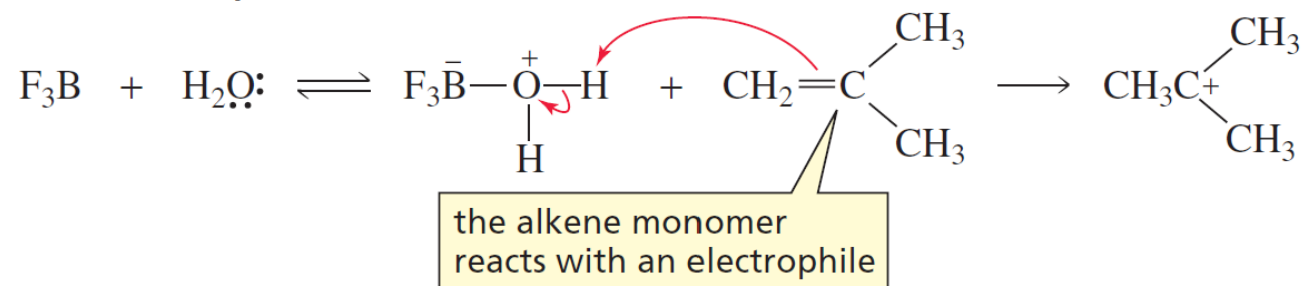


## Cationic Polymerization

In cationic polymerization, the initiator is an electrophile (generally a proton) that adds to the monomer, causing it to become a carbocation. The initiator cannot be an acid such as HCl because  $\text{Cl}^-$  will be able to react with the carbocation. Thus, the initiator most often used in cationic polymerization is a Lewis acid, such as  $\text{BF}_3$ , together with a proton-donating Lewis base, such as water. Notice that the reaction follows the rule that governs electrophilic addition reactions—that is, the electrophile (the initiator) adds to the  $sp^2$  carbon bonded to the most hydrogens

### MECHANISM FOR CATIONIC POLYMERIZATION

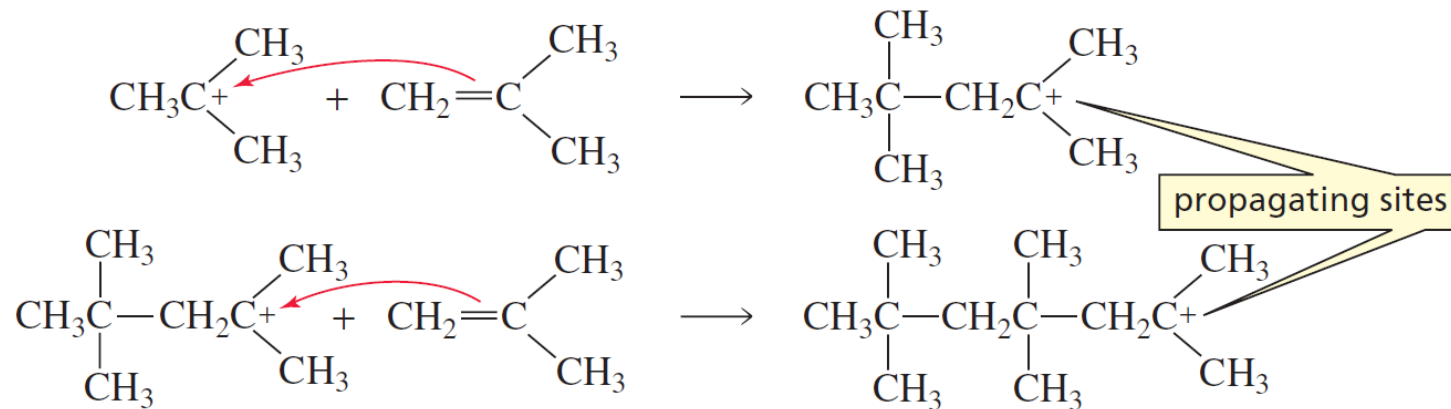
initiation steps



## Cationic Polymerization

- The cation formed in the initiation step reacts with a second monomer, forming a new cation that reacts in turn with a third monomer. As each subsequent monomer adds to the chain, the new positively charged propagating site is at the end of the most recently added unit.

propagating steps



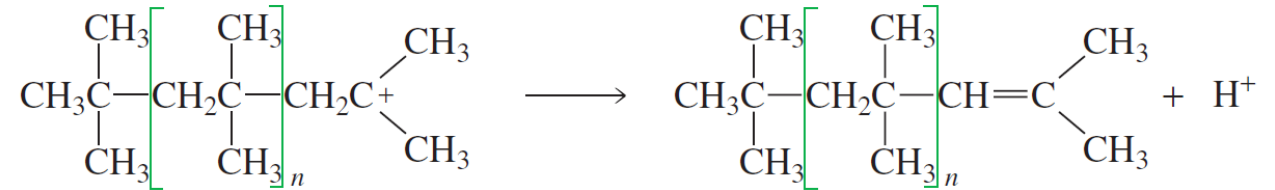
## Cationic Polymerization

Cationic polymerization can be terminated by

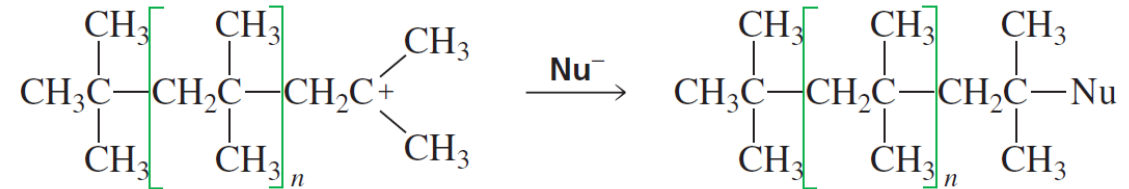
- loss of a proton;
- addition of a nucleophile to the propagating site;
- a chain-transfer reaction with the solvent (XY).

### termination steps

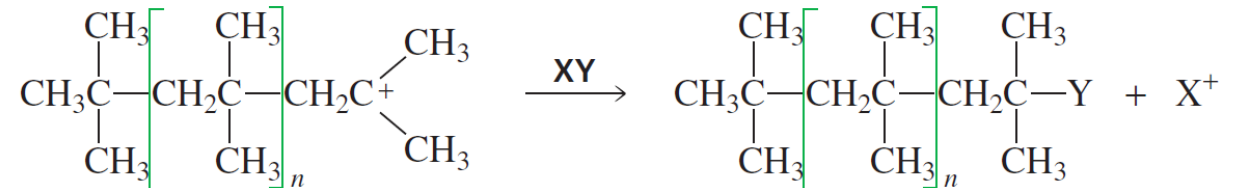
#### loss of a proton



#### reaction with a nucleophile

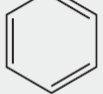


#### chain-transfer reaction with the solvent



## Cationic Polymerization

Monomers that are best able to undergo polymerization by a cationic mechanism are those with substituents that can stabilize the positive charge at the propagating site either by hyperconjugation (the first compound in Table below) or by donating electrons by resonance (the other two compounds in the table)

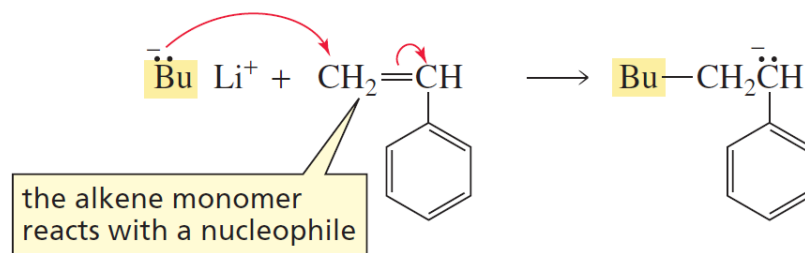
| Table Examples of Alkenes That Undergo Cationic Polymerization  |                                                                      |                                                                                                                                |
|-----------------------------------------------------------------|----------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| $\text{CH}_2=\text{CCH}_3$<br> <br>$\text{CH}_3$<br>isobutylene | $\text{CH}_2=\text{CH}$<br> <br>$\text{OCH}_3$<br>methyl vinyl ether | $\text{CH}_2=\text{CH}$<br> <br><br>styrene |

# Anionic Polymerization

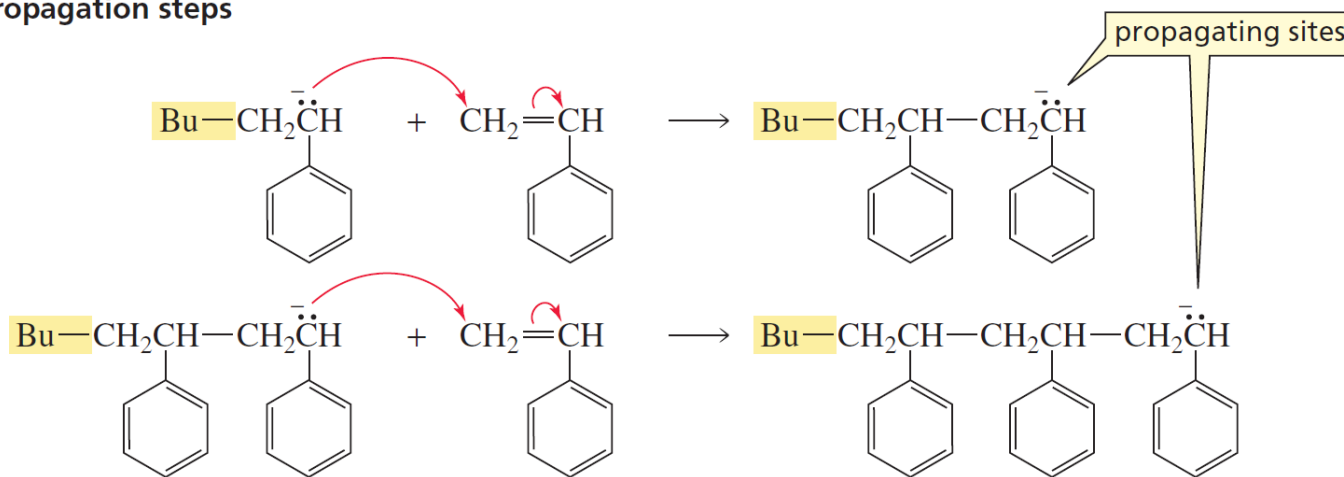
In anionic polymerization, the initiator is a nucleophile that reacts with the monomer to form a propagating site that is an anion. Nucleophilic attack on an alkene does not occur readily because alkenes are themselves electron rich. Therefore, the initiator must be a very good nucleophile, such as sodium amide or butyllithium, and the alkene must contain a substituent that can withdraw electrons by resonance, which will decrease the electron density of the double bond.

## MECHANISM FOR ANIONIC POLYMERIZATION

### initiation step



### propagation steps



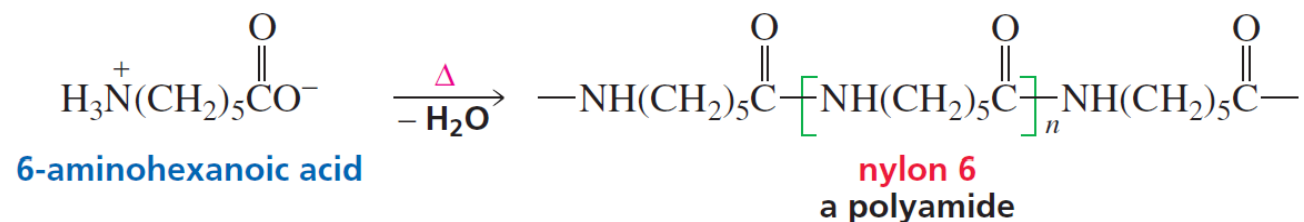
## Anionic Polymerization

The chain can be terminated by reaction with an impurity in the reaction mixture. If all impurities are rigorously excluded, chain propagation will continue until all the monomer has been consumed. At this point, the propagating site will still be active, so the polymerization reaction will continue if more monomer is added to the system. Such nonterminated chains are called **living polymers** because the chains remain active until they are terminated (“killed”).

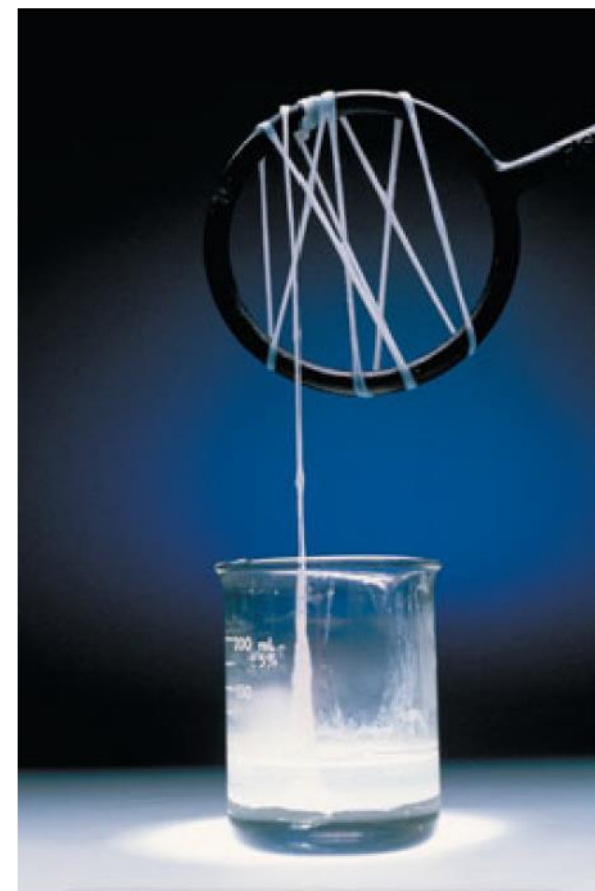
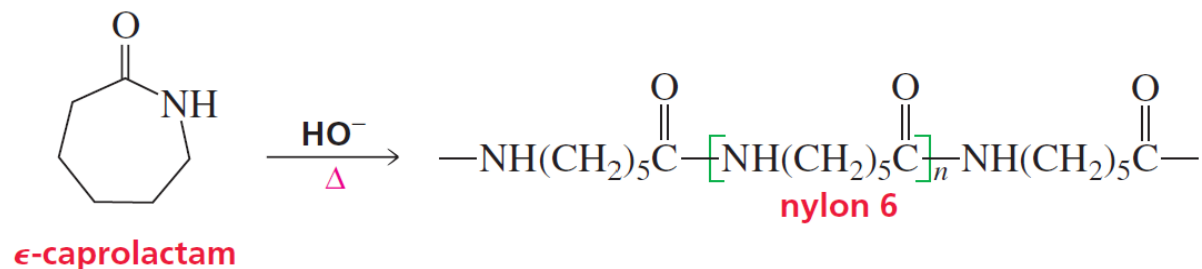
| Table                                                                                                    | Examples of Alkenes That Undergo Anionic Polymerization                                                                    |                                                                                                  |  |
|----------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|--|
| $\begin{array}{c} \text{CH}_2=\text{CH} \\   \\ \text{CH} \\    \\ \text{O} \end{array}$ <p>acrolein</p> | $\begin{array}{c} \text{CH}_2=\text{CCH}_3 \\   \\ \text{COCH}_3 \\    \\ \text{O} \end{array}$ <p>methyl methacrylate</p> | $\begin{array}{c} \text{CH}_2=\text{CH} \\   \\ \text{C}_6\text{H}_5 \end{array}$ <p>styrene</p> |  |

## Polyamides

Nylon 6 is an example of a step-growth polymer formed from a monomer with two different functional groups. The carboxylic acid group of one monomer reacts with the amino group of another monomer to form an amide. Thus, nylon is a **polyamide**. This particular nylon is called nylon 6 because it is formed from the polymerization of 6-aminohexanoic acid, a compound that contains six carbons.



The starting material for the commercial synthesis of nylon 6 is  $\epsilon$ -caprolactam. The lactam is opened by a base.

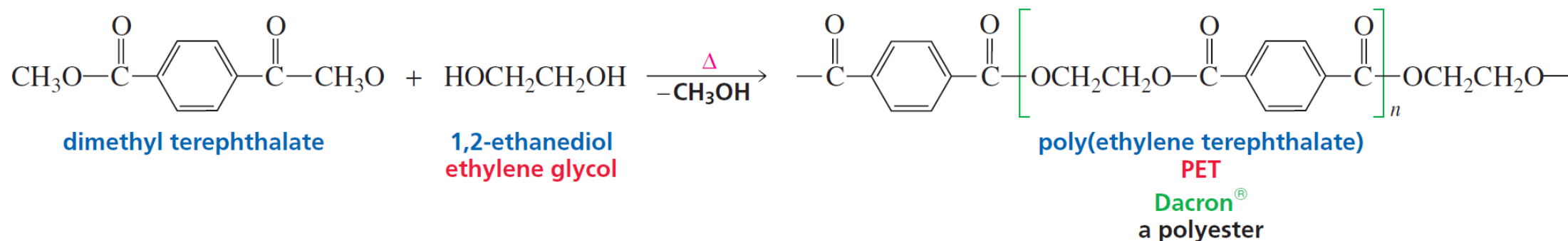


Nylon is pulled from a beaker of adipoyl chloride and 1,6-hexanediamine.

## CLASSES OF STEP-GROWTH POLYMERS

### Polyesters

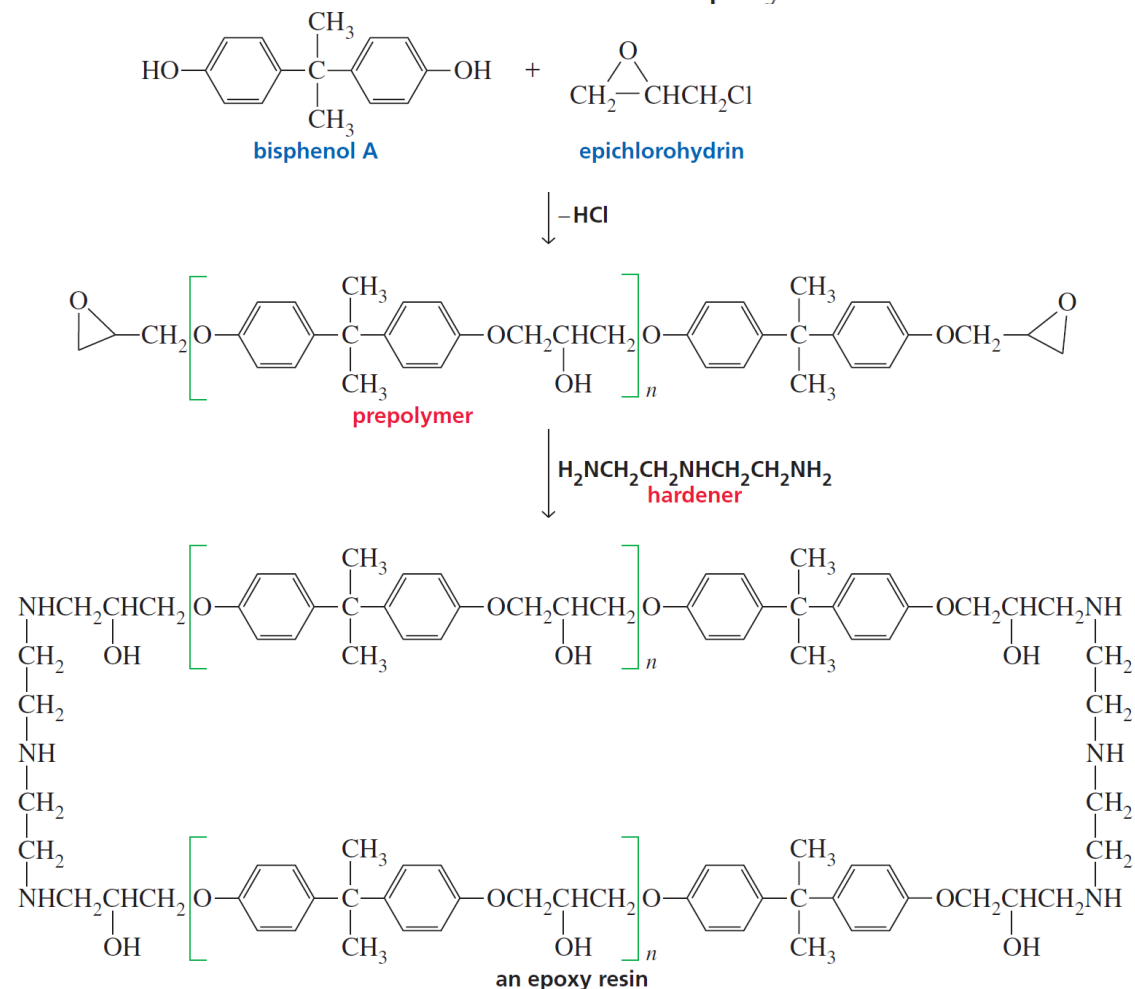
Dacron is the most common of the group of step-growth polymers known as **polyesters**—polymers containing many ester groups. Polyesters are used for clothing and are responsible for the wrinkle-resistant behavior of many fabrics. Dacron is made by the transesterification of dimethyl terephthalate with ethylene glycol. The durability and moisture resistance of this polymer contribute to its “wash-and-wear” characteristics. Because PET is lightweight, it is also used for transparent soft drink bottles.



# CLASSES OF STEP-GROWTH POLYMERS

## Epoxy Resins

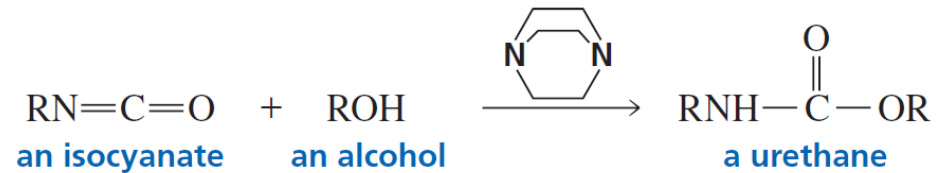
**Epoxy resins** are the strongest adhesives known; they are extensively cross-linked systems. They can adhere to almost any surface and are resistant to solvents and to extremes of temperature. Epoxy cement is purchased as a kit consisting of a low-molecular-weight *prepolymer* (the most common is a copolymer of bisphenol A and epichlorohydrin) and a *hardener* that react when mixed to form a cross-linked polymer.



## CLASSES OF STEP-GROWTH POLYMERS

# Polyurethanes

A **urethane**—also called a carbamate—is a compound that has an OR group and an NHR group bonded to the same carbonyl carbon. Urethanes can be prepared by treating an isocyanate with an alcohol, in the presence of a catalyst such as a tertiary amine.



One of the most common **polyurethanes**—polymers that contain urethane groups—is prepared by the polymerization of toluene-2,6-diisocyanate and ethylene glycol.

